INDIVIDUAL WEEKLY PROJECT REPORTS

Student: Annalakshmi R

Role: Data Preprocessing and Exploratory Data Analysis (EDA)
Project Title: Flight Delay Analysis and Visualization

WEEK 7 – Correlation Analysis and PCA Support

During the seventh week, I focused on supporting the statistical analysis of the flight delay dataset. The team examined relationships between numerical variables using correlation analysis and explored Principal Component Analysis (PCA) as a dimensionality reduction technique.

I concentrated on understanding the relationship between departure delay and arrival delay. The analysis indicated a strong positive correlation of approximately 0.971 between these two variables. This suggested that flights experiencing greater departure delays generally also experienced greater arrival delays.

I also familiarized myself with the purpose of PCA. PCA is a statistical technique used to transform a set of correlated variables

into a smaller number of principal components while retaining as much of the original variation as possible.

My contribution involved reviewing the numerical variables used in the analysis and understanding how the results connected with the earlier EDA findings. I also coordinated with the team members responsible for documentation and visualization to share relevant observations.

I understood that statistical analysis could provide additional information beyond basic descriptive statistics and help explain relationships within the dataset.

Outcome and Learning:

This week improved my understanding of correlation analysis and PCA. I learned how statistical methods can be used to identify relationships and summarize information in a large dataset.